

DOMAIN MAPPER

*Active Directory Enumeration & Exploitation
Framework*

Cybersecurity Bash Automation Project

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Project Documentation
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Scanning • Enumeration • Exploitation • Reporting

Project Overview

Domain Mapper is a Bash automation framework designed for scanning, enumeration, and exploitation in Active Directory environments. The script combines multiple cybersecurity tools such as Nmap, CrackMapExec, NetExec, John the Ripper, and Impacket into one workflow.

1. Scanning Mode

Basic Scan: Uses Nmap with the -Pn flag to identify active systems without host discovery.

Intermediate Scan: Scans all 65535 TCP ports using the -p- option.

Advanced Scan: Adds UDP scanning (-sU) for deeper network analysis.

2. Enumeration Mode

Basic Enumeration: Detects services, identifies the Domain Controller IP, and discovers the DHCP server.

Intermediate Enumeration: Enumerates important services such as SMB, LDAP, RDP, FTP, SSH, and WinRM. Also uses NSE scripts for domain analysis.

Advanced Enumeration: Uses Active Directory credentials to extract users, groups, shares, password policies, disabled accounts, and Domain Admin information.

3. Exploitation Mode

Basic Exploitation: Runs Nmap vulnerability scanning scripts against the target domain.

Intermediate Exploitation: Performs password spraying attacks using CrackMapExec.

Advanced Exploitation: Extracts Kerberos AS-REP tickets using Impacket and attempts to crack them with John the Ripper.

4. Results & Output

The script automatically saves all outputs into organized files and generates a final PDF report using Enscript and ps2pdf.

1 In the first part of the script, it will run a full tool check making sure that the user has all the nesscery tools in order to run any type of option in the script, if the user is missing any script it will run an install for the missing one.

```
#####  
[O]nline[M]ap[O]  
#####  
Checking for required tools...  
[√] nmap is installed  
[√] crackmapexec is installed  
[√] netexec is installed  
[√] cowsay is installed  
[√] enscript is installed  
[√] ps2pdf is installed  
[√] impacket-GetNPUsers is installed  
[√] john is installed
```

2 On the next part after the tool check the user will be asked a series of questions to get the info needed for the script

```
Enter Network range for scanning: 192.168.1.0/24
Enter the Domain Controller name: netsec.local
Enter AD username (press enter to skip): Administrator
Enter AD password (press enter to skip): 12345678
Specify a name of a new folder to save all the data: project34
Specify the password list to use (press enter to default): pass.txt
```

Note if the user doesn't give a pass list it will use the rockyou pass list the user is not force to enter all info it will effect what the script can do in the future.

3 Now the user will get the option to choose which scan type he would like the script to operate , basic ,intermediate or advanced. Each type will scan the domain and collect data on it

Choose scan type
B for basic scan, I for intermediate, A for advanced, N to skip

After choosing a scan type the user will be notified that the scan type he choose was started like shown here

b

< Basic scanning : >



4 After the scan finishes the user will get the option to choose which enumeration type he would like the script to perform just like the scan type

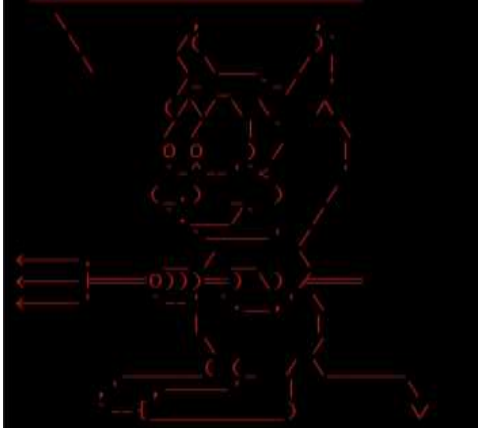
Choose Enumeration type
B for basic, I for intermediate, A for advanced, N to skip

Note if the user choose to skip enum part then the script wont be able to continue to the exploit part and will exit

```
n
Skipping enum.
Domain IP not found. Please run enumeration first.
```

If the user ran any enum type Same like the scan choice the script will tell the user that the enum he chose started

```
a
< Starting Advanced enum: >
```



Then the user will get some data on the screen (most data gonna be in the file in the end) the data output will change it depends on the enumeration type the photo shown next is the advanced enum option

```
The Domain IP is 192.168.1.100
the DHCP server IP is 192.168.1.1
The following IPs has the port 21 on
192.168.1.100

The following IPs has the port 22 on

The following IPs has the port 445 on
192.168.1.100

The following IPs has the port 5985 on
192.168.1.100

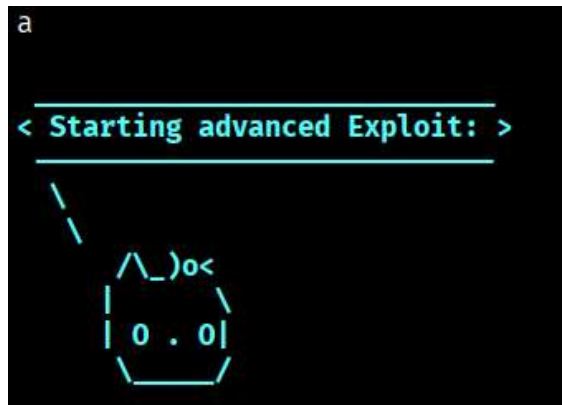
The following IPs has the port 389 on
192.168.1.100

The following IPs has the port 3389 on
```

5 The next part will be the exploit ! , ofc here as well the user will get the option to choose which diffrence exploit he would like

```
Choose Exploit type
B for basic Exploitation, I for intermediate, A for advanced, N to skip
```

As well like the other options the script will tell the user when the action started



As well in the advanced exploit mode the script will attempt to brute force the user hashes if not succsefuel the following wlll show

```
Didn't manage to brute force the hashes
```

After the exploit type the script will make a big pdf file with all the data inside it

```
pdf file created name output.pdf in the project99 directory
```



6 In the end the user will receive the file in addition to the pdf file with all the data inside of it (of course the data will change depends on the options chosen by the user)

The file

